For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

MATHEMATICS — 9164/2

PAPER 2

NOVEMBER 2009 SESSION — 3 hours

Additional materials:
- Answer paper
- Graph paper
- List of Formulae

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

There is no restriction on the number of questions which you may attempt.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given to the nearest degree, and in other cases it should be given correct to 2 significant figures.

If a numerical value for g is necessary, take $g = 9.81\ ms^{-2}$.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is 120.

Within each section of the paper, questions are printed in the order of their mark allocations and candidates are advised, within each section, to attempt questions sequentially.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

This question paper consists of 6 printed pages and 2 blank pages.

Section (a): Pure Mathematics

1 Prove by induction that $5^{2n} - 3^{2n}$ is a multiple of 8 for $n \in Z^+$. [7]

2 Given that $z = \cos\theta + i\sin\theta$. Show that $z - \dfrac{1}{z} = 2i\sin\theta$. [3]

Hence express $\sin^4\theta$ in terms of $\cos^4\theta$ and $\cos 2\theta$ using De Moivre's theorem. [4]

3 **(a)** Expand $\left(1 - \dfrac{x}{n}\right)^n$ up to and including the term in x^3 simplifying the coefficients. [3]

(b) If the term in x^3 is $\dfrac{7}{8}$ when $x = -2$ and n is a positive integer, find the value of n. [5]

4 In 2005 a community of 10 000 people planned an extension to its water facilities. Assume that the birth rate and death rate are 55 persons per 1 000 and 14 persons per 1 000 respectively, and that there are no other changes in population. The population is P persons at time t years after 2005.

The rate of increase of the population is given by the product of the net-rate of increase and the population at any time.

Show that the differential equation which models this situation can be written as

$$\frac{dP}{dt} = \frac{41}{1000}P.$$

Solve this equation and determine the predicted population for the year 2020. [8]

5 Given that $A = \begin{pmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 3 & 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 2 & 1 \\ 1 & 3 & 0 \end{pmatrix}$

(i) Find AB. [3]

(ii) Given that $AB\begin{pmatrix} -6 & -18 & 15 \\ 4 & 8 & -7 \\ 4 & 14 & -10 \end{pmatrix}$ can be expressed in the form

$\begin{pmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{pmatrix}$. Find the value of the constant a. [2]

(iii) It is also given that $AB\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$, find x, y, z. [4]

6 **(a)** The line l passes through the points Q(3, 1, –2) and R(2, 7, –4). Find

(i) the cartesian equation of the line l, [4]

(ii) the point at which l intersects the plane $x - y$. [3]

(b) The planes P_1, P_2 and P_3 have the equations

$2x - 3y - z - 5 = 0$, $-6x + 9y + 3z + 2 = 0$ and
$3x + 2y - 6z + 10 = 0$ respectively.

(i) Show that planes P_1 and P_2 are parallel to each other. [2]

(ii) The plane P_4 passes through P(5, –2, 4) and is parallel to P_3.
Find its equation. [4]

(iii) Find the perpendicular distance of P(5, –2, 4) from the plane P_1. [3]

7 **(a)** Express $4(\sqrt{3} - i)$ in the form $re^{i\theta}$ where $r > 0$ and $-\pi < \theta \leq \pi$. [3]

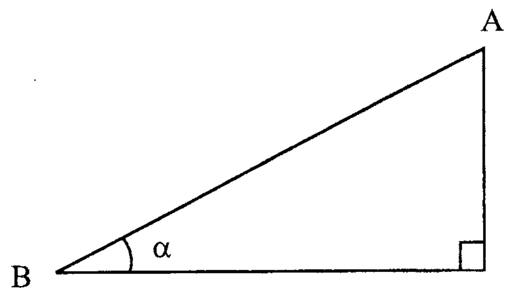
(b) Given that $x_1 = 1 + 2i$ is a root of the equation
$x^4 - 4x^3 - 6x^2 + 20x - 75 = 0$, find the other three roots. [5]

(c) Express $\sin 5\theta$ in terms of powers of $\sin\theta$ and hence show that
$\sin 5\theta - 5\sin\theta = 16\sin^5\theta - 20\sin^3\theta$.

Find $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \left(16\sin^5\theta - 20\sin^3\right) d\theta$, giving your answer in exact form. [9]

Section (b): Mechanics

8 A block of mass 3.5 kg is released from rest at point A on a plane inclined at angle α to the horizontal, where $\tan\alpha = \frac{3}{4}$. The block slides down until it reaches point B at the base of the plane (see diagram).

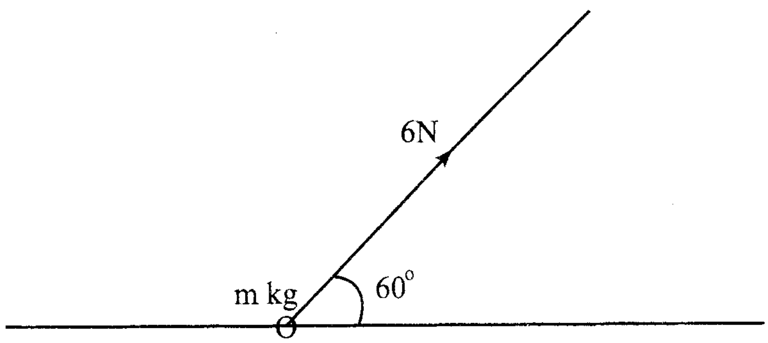


Given that the coefficient of friction between the block and the plane is $\frac{1}{4}$, calculate

(i) the velocity of the block at the instant it reaches point B, [5]

(ii) the time it takes the block to slide from point A to point B. [2]

9



The diagram above shows a ring of mass m kg accelerating at 1 ms^{-2} along a rough horizontal wire. The accelerating force of 6 N is at an angle of 60° in the same vertical plane with the wire.

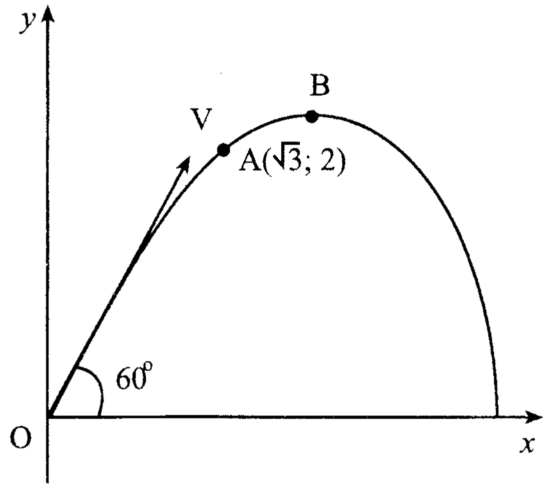
The coefficient of friction between the ring and the wire is $\frac{1}{4}$. Find in terms of m and, or g, the exact value of

(i) the normal reaction between the ring and the wire, [2]

(ii) the fiction force. [2]

Hence find the value of m, giving your answer to 3 decimal places. [3]

10 A particle is projected from a point O on the ground with a speed of V ms^{-1} at an angle of 60° to the horizontal and passes through the points A and B, where $A(\sqrt{3}\,;2)$ is a point before it reaches its maximum height above O at B (see diagram).



(a) Express V^2 in terms of g. [4]

(b) Find the angle that AB makes with the horizontal. [6]

Section (c): Statistics

11 The owners of a motel in Mutare have noticed that in the long run 40% of the people who stop and inquire about a room for the night, actually book a room.

How many inquiries must the owners answer to be 99% sure of at least one booking? [5]

12 **(a)** **(i)** Give **two** advantages of using stem and leaf diagrams in analysing data. [2]

(ii) Describe with an example a statistical situation in which it would be appropriate to use the mode as a measure of central tendency. [1]

(b) The mean and standard deviation of the masses of a group of adult males are 65 kg and 10 kg respectively. Males are considered overweight if they are in the top 5% of the group by mass. Assuming that the masses of this group are normally distributed, find the least mass to be considered overweight. [3]

13 Three tickets for a musical show are sent to a high school musical club. Fifteen girls and ten boys would like a ticket. If the three people to receive a ticket are chosen at random, find the probability that will there be

(i) exactly 2 boys, [3]

(ii) at least 2 girls. [3]

14 The duration X minutes of a telephone call by a school head to the Provincial Education Director, is a continuous random variable with a probability density function defined by

$$f(x) = \begin{cases} x^{-2}, & x \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Given that a call has already lasted 5 minutes, find the conditional probability that its total duration will be less than 7 minutes. [7]